



Investor Quest – Developer Handoff

Schools Demo Prototype

Technical handoff for developers polishing and extending the
Schools presenter demo and Business Island experience.

<https://investor-quest.vercel.app/schools/demo>

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Investor Quest — Developer Handoff

Prototype status: Schools presenter demo + **Business Island** are the polished priority. The other three islands (Forces/Risks, Financials, Management) exist in the wider product concept but are **not** at the same level of refinement.

Your main job: Polish and extend the existing experience — onboarding, map, Business Island, quest cards, quizzes, XP/progression, profile, mobile/desktop quality — without needing to rebuild the core architecture.

Live Demo

Environment	URL
Production (Vercel)	https://investor-quest.vercel.app/schools/demo
Business island (prod)	https://investor-quest.vercel.app/schools/business → redirects into zoomed map hub
Schools map (prod)	https://investor-quest.vercel.app/schools/map
Schools profile (prod)	https://investor-quest.vercel.app/schools/profile
Main quest map (prod)	https://investor-quest.vercel.app/map

Matching local URLs (same flow)

URL	Notes
http://localhost:3003/schools/demo	Local presenter entry (sidebar Schools Live Demo launches here)
http://localhost:3003/schools/map	Prodigy overworld map
http://localhost:3003/schools/business	Legacy route → map zoom hub
http://localhost:3003/schools/business/what-they-do	First Business quest (Schools context)
http://localhost:3003/schools/profile	Schools profile dashboard
http://localhost:3003/admin/game-health	Dev ops / route probes (needs env)

Port: Always **3003** (`npm run dev`). Bookmarks and Capacitor local sync assume this port.

Deploy: Pushes to `main` on GitHub (`collo1978/investor-quest`) auto-deploy to Vercel. Production origin is hard-coded in `src/lib/schools/schoolsDemoPwa.ts` as `https://investor-quest.vercel.app` . Capacitor wrappers also point at `/schools/demo` on that host (`capacitor.config.ts` , `docs/CAPACITOR.md`).

1. Tech stack & tooling

Area	Choice	Notes
Framework	Next.js 15.4 (App Router)	<code>src/app/**/page.tsx</code> routing
Language	TypeScript 5.8	Strict types in <code>src/data/quests/types.ts</code> , engine layer
React	19.1	Client components marked <code>"use client"</code>
Styling	Tailwind CSS 4 + large custom CSS	<code>postcss.config.mjs</code> → <code>@tailwindcss/postcss</code> . Most Schools/Business polish lives in <code>src/app/globals.css</code> (<code>iq-*</code> classes). Utility merge via <code>clsx</code> + <code>tailwind-merge</code> (<code>src/lib/utls.ts</code>)
Animation	Framer Motion 12 (<code>framer-motion</code> , <code>motion</code>)	Quest transitions, map zoom, quiz micro-interactions
Confetti	canvas-confetti	Correct-answer celebrations (<code>src/lib/quests/fireQuizCorrectConfetti.ts</code>)
3D (limited)	three , @react-three/fiber , cobe	Globe/visual flourishes; not core to Schools demo
UI primitives	Radix UI + many generated components under <code>src/components/ui/</code>	Accordion, scroll-area, buttons, magic cards, etc. — treat as shadcn-style building blocks
State	React Context + pure reducer	<code>GameProvider</code> + <code>src/engine/progression/reducer.ts</code> — not Redux/Zustand
Persistence	localStorage (primary)	<code>src/engine/persistence/localStorageStore.ts</code> , key <code>investor-quest::state</code>
Backend (optional)	Supabase (<code>@supabase/supabase-js</code> , <code>@supabase/ssr</code>)	Quest card generation pipeline, admin — Schools demo works offline with hand-authored content
Mobile shell	Capacitor 8	Wraps Vercel <code>/schools/demo</code> ; see <code>docs/CAPACITOR.md</code>
Node	>= 20.9.0	<code>package.json</code> <code>engines</code>

Key dependency versions (`package.json`)

```

next ^15.4.0
react ^19.1.0
typescript ^5.8.3
tailwindcss ^4.1.10
framer-motion ^12.42.0
@supabase/supabase-js ^2.105.4
@capacitor/core ^8.3.4
canvas-confetti ^1.9.4

```

AI-generated code & assets (obvious patterns)

Code likely AI-assisted or pipeline-generated:

- SEC → OpenAI quest card pipeline (`src/lib/sec/`, `src/lib/ai/`, admin `/admin/quest-generation`)
- Large batch of `src/components/ui/*` effect components (animated beams, neon cards, etc.)
- Quest templates with `plainEnglishAnswer: null` awaiting generation (`src/data/quests/templates/*.ts`)

- Dynamic quiz transforms (`src/lib/quests/dynamicQuizEngine.ts`) — turns MC into T/F, fill-blank, odd-one-out, order, scenario per attempt

Assets that appear AI-generated / painted raster art:

- Fullscreen onboarding posters in `public/logos/` (`opening-new-screen.png` , `schools-sounds-like-you.png` , `screen2-onboarding.png` ... `screen5-onboarding.png` , `choose-your-investor-armor.png` , mission briefs)
- Island hub paintings in `public/screens/` (`biz-quest.webp` , `financial-quest.webp` , `forces-quest.webp` , `management-quest.webp`)
- Schools identity avatars `public/images/schools/avatars/{alex,zoe,jayden,...}.png`
- Schools map raster `public/logos/new-map.png` , `latest-map-schools.png` (preview/legacy — **live map uses SVG**)
- **Dev artifacts at repo root:** `tmp-zone-*.jpg` — hotspot calibration images; **not shipped**, safe to delete

Coded (non-AI) visuals:

- **Prodigy map** — pure SVG (`src/components/schools/prodigyMap/ProdigyMapWorld.tsx` , `ProdigyMapLandmarks.tsx`)
- **Schools Business hub hydration scene** — coded SVG (`src/components/schools/SchoolsBusinessHubCodedScene.tsx`)
- Company logos in `public/logos/companies/*.svg` — simple geometric marks (some extras are placeholders)

2. Project structure

```
investor-quest/
├── src/
│   ├── app/                # Next.js App Router pages
│   ├── components/         # React UI (game + schools + business + ui/)
│   ├── data/               # Pure content: companies, pillars, quests, progression
│   ├── engine/             # Pure game logic: reducer, XP, badges, persistence
│   ├── hooks/              # React hooks bridging UI ↔ engine/framework storage
│   ├── lib/                # Feature Logic (schools, business, quests, profile, ...)
│   ├── platform/           # Quest catalog registry (demo vs Supabase)
│   ├── sec/                # SEC fetch/parse stubs
│   ├── ui/                 # Shared effects (confetti, Level-up FX)
│   └── middleware.ts        # Pathname header for SSR layout
├── public/                 # Static assets (Logos, screens, sounds, schools PWA)
├── scripts/                # Dev server, quest regen, game-health cron, doc gen
├── supabase/migrations/    # DB schema (game health, quest cards) – optional locally
├── docs/                   # CAPACITOR.md, etc.
├── capacitor.config.ts     # Mobile app entry → Vercel /schools/demo
├── HANDOFF.md              # This file
└── README.md               # Local dev quick start
```

Directory purposes (important ones)

Path	Purpose
<code>src/app/schools/</code>	Schools demo routes — onboarding, map, business, profile, preview lab
<code>src/app/schools/demo/</code>	Presenter PWA mirror of <code>/schools/*</code> with <code>/schools/demo</code> prefix
<code>src/app/schools/preview/</code>	Experimental map/mission-brief style explorations (not live tour)
<code>src/app/business/</code>	Canonical Business Island quest routes (<code>/business/[questSlug]</code>)
<code>src/app/map/</code>	Main cinematic quest map (<code>desktop-map.png</code>) — separate from Schools Prodigy map
<code>src/app/profile/</code>	Profile hub (<code>InvestorProfileHub</code>)
<code>src/app/admin/</code>	Ops: quest generation, game health, branding, analytics
<code>src/components/schools/</code>	Schools-only screens (map, onboarding, mission brief, celebrations)
<code>src/components/business/</code>	Business hub markers, checklist, investor framework/quality UI
<code>src/components/QuestQuizPanel.tsx</code>	Core quiz runner (all islands)
<code>src/components/QuizQuestionView.tsx</code>	Per-question renderers (MC, T/F, fill-blank, order, ...)
<code>src/components/BusinessIslandQuestReading.tsx</code>	Main Business quest reading + quiz + checklist orchestration
<code>src/components/profile/InvestorProfileDashboard.tsx</code>	Live-coded profile dashboard (not a static image)
<code>src/data/companies/</code>	6 playable companies (AAPL, MSFT, TSLA, NVDA, NKE, SPOT)
<code>src/data/quests/templates/business.ts</code>	7 Business quest templates + card structure
<code>src/data/quests/businessQuestQuizzes.ts</code>	Per-quest quiz definitions (MC base content)
<code>src/lib/business/</code>	Investor checklist, evidence, section quizzes, hub cards
<code>src/lib/schools/</code>	Demo story mode, navigation, rewards, map config, PWA
<code>src/lib/quests/dynamicQuizEngine.ts</code>	Section quiz format variety (5 distinct mini-game layouts)
<code>src/engine/progression/</code>	Reducer, XP, badges, unlocks, persistence migration
<code>src/lib/screenAssetUrls.ts</code>	Central registry for hub/map image paths + cache-bust revisions

Files to open first (Business Island + demo)

1. `src/lib/schools/launchSchoolsProductionDemo.ts` — how presenter demo starts
2. `src/lib/schools/schoolsDemoStoryMode.ts` — scripted tour steps + routes
3. `src/app/schools/map/SchoolsMapPageClient.tsx` — map + envelope brief + Business zoom
4. `src/components/business/hub/SchoolsBusinessHubIslandLayout.tsx` — hub gameplay UI
5. `src/components/BusinessIslandQuestReading.tsx` — quest cards → quiz → quality check → checklist
6. `src/components/QuestQuizPanel.tsx` + `src/components/QuizQuestionView.tsx` — quiz UX

7. `src/lib/business/businessInvestorFramework.ts` + `businessChecklistSectionQuizzes.ts` — checklist content
8. `src/components/profile/InvestorProfileDashboard.tsx` — profile polish target
9. `src/components/GameProvider.tsx` + `src/engine/progression/reducer.ts` — XP/progression

See also `src/ARCHITECTURE.md` for engine layer overview (slightly older but accurate on reducer/store split).

3. Run it locally

Install

```
# Node >= 20.9.0
npm install
```

Dev server

```
npm run dev
# → http://localhost:3003 (Ctrl+Shift+B in Cursor also works)
```

Other scripts:

Command	Use
<code>npm run dev:fresh</code>	Clear <code>.next</code> cache + restart
<code>npm run dev:kill-port</code>	Free port 3003
<code>npm run build</code>	Production build (Vercel uses this)
<code>npm run lint</code>	ESLint
<code>npx tsc --noEmit</code>	Typecheck without emit

Start the Schools demo locally

1. Run `npm run dev`
2. Open app with sidebar → click **Schools Live Demo** (calls `launchSchoolsProductionDemo` in `src/components/AppShell.tsx`)
3. Or navigate directly to <http://localhost:3003/schools/demo>

This clears prior saves, seeds demo game state, activates story mode, and hard-navigates to the mission invitation screen.

Environment variables

Needs verification: Full list lives in `.env.local` (not committed). The Schools **presenter demo does not require** Supabase or OpenAI to run — it uses hand-authored quest content and `localStorage`.

Variables referenced in project docs / code paths (set in `.env.local` for full platform features):

Variable	Purpose	Where used
<code>NEXT_PUBLIC_APP_URL</code>	Canonical app origin fallback	<code>src/lib/schools/schoolsDemoPwa.ts</code>
<code>NEXT_PUBLIC_SUPABASE_URL</code>	Supabase project URL	Admin quest generation, company cards
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Supabase anon key	Client fetches
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Server-side Supabase	API routes / admin
<code>OPENAI_API_KEY</code>	Quest card AI generation	SEC pipeline, <code>/admin/quest-generation</code>
<code>QUEST_FAST_DEV</code> / <code>NEXT_PUBLIC_QUEST_FAST_DEV</code>	Skip slow jargon loops in dev	Quest pipeline (see <code>.cursor/rules/dev-server-port.mdc</code>)
<code>GAME_HEALTH_BASE_URL</code>	Health probe target	<code>http://localhost:3003</code> locally
<code>GAME_HEALTH_CRON_SECRET</code>	Cron auth for health checks	Optional
<code>RESEND_API_KEY</code> , <code>GAME_HEALTH_ALERT_EMAIL</code>	Email alerts	Optional ops

For handoff polish work: You can ignore most of these. Only add Supabase/OpenAI if you are extending the AI quest pipeline.

External services (optional)

- **Vercel** — hosting (GitHub `main` → production)
- **Supabase** — generated quest card storage, game health migration (`supabase/migrations/`)
- **OpenAI** — quest content generation (admin)
- **Resend** — optional game-health email alerts

4. Live deployment

Item	Value
Production URL	https://investor-quest.vercel.app/schools/demo
Git remote	<code>https://github.com/collo1978/investor-quest.git</code>
Deploy branch	<code>main</code>
PWA SW version	<code>src/lib/schools/schoolsDemoPwa.ts</code> → <code>SCHOOLS_DEMO_SW_VERSION</code> (bump when shipping Schools fixes)

If the Vercel URL ever changes, update:

- `src/lib/schools/schoolsDemoPwa.ts` (`SCHOOLS_DEMO_PRODUCTION_ORIGIN`)
- `capacitor.config.ts`
- `docs/CAPACITOR.md`
- `README.md` (Live production section)

- iOS/Android `capacitor.config.json` allowlists

5. Architecture & data flow

High-level mutation flow

```
UI (useGame().actions.*)
  → GameProvider.dispatch
  → applyAction → reducer.ts (pure)
  → new GameState + RewardEvent[]
  → ProgressionStore.save() (localStorage)
  → Toasts / FX (ToastHost, QuestCompletionFx, ConvictionQueueHost)
```

Game state

Concern	File(s)
Canonical state shape	<code>src/engine/progression/state.ts</code> (<code>GameState</code> , <code>STATE_VERSION = 13</code> , <code>STORAGE_KEY</code>)
All mutations	<code>src/engine/progression/reducer.ts</code>
React adapter	<code>src/components/GameProvider.tsx</code>
Load/save/migrate	<code>src/engine/progression/persistence.ts</code>
localStorage adapter	<code>src/engine/persistence/localStorageStore.ts</code>
Hydration gate	<code>src/lib/gameState/persistenceGate.ts</code>
Cross-tab merge	<code>src/lib/gameState/mergeLoadedGameState.ts</code>

Important: `markQuestRead` tracks reading progress (no XP). `completeQuest` on quiz pass awards XP and marks quest complete.

XP & levels

Concern	File(s)
Engine XP constants	<code>src/engine/progression/xpEconomy.ts</code> — section quiz 50, perfect bonus 25, island 200, 10-K 300
Level ladder	<code>src/data/progression/investorLadder.ts</code>
Level math	<code>src/engine/progression/xp.ts</code>
Schools micro-XP	<code>src/lib/schools/schoolsQuestRewardFlow.ts</code> — 10 XP/correct, 75 quest-clear bonus (UI layer; separate from engine constants)
XP UI	<code>src/components/LevelBar.tsx</code> , profile hero in <code>InvestorProfileDashboard.tsx</code>

Schools quest reward path: Read cards → `QuestQuizPanel` → `SchoolsQuizPassResultsCelebration` (micro-XP recap) → Investor Quality Check → `SchoolsQuestQuizCompletionFlow` (quest XP + skill copy) → back to hub.

Unlocks & pillars

Concern	File(s)
Pillar metadata	<code>src/data/pillars/index.ts</code>
Unlock policy	<code>src/engine/progression/unlocks.ts</code> , <code>pillarUnlockPolicy.ts</code>
Business quest chain	Templates set <code>unlockRequirements</code> in <code>src/data/quests/templates/business.ts</code>

Schools demo starts with Business pillar active; other islands are reachable via menu but less polished.

Badges

Concern	File(s)
Badge definitions	<code>src/engine/progression/badges.ts</code>
Detection on state change	<code>detectNewBadges()</code> called from reducer / <code>useInvestorProfileSnapshot</code>
Checklist badges	Awarded when investor framework milestones hit

Quiz completion

Flow	File(s)
Standard island quiz	<code>QuestQuizPanel.tsx</code> → <code>actions.completeQuest(pillarId, slug, { quizPerfect })</code>
Section checklist quiz (5 questions)	<code>BusinessChecklistSectionQuizFlow.tsx</code> → <code>buildDynamicQuizConfig()</code> → <code>QuestQuizPanel</code> → <code>markSectionQuizPassed()</code> in <code>businessInvestorFrameworkStorage.ts</code>
Dynamic format variety	<code>src/lib/quests/dynamicQuizEngine.ts</code> , <code>quizLayoutProfiles.ts</code>
Answer rendering	<code>src/components/QuizQuestionView.tsx</code>

Conviction / confidence

Concern	File(s)
Post-island conviction queue	<code>src/components/conviction/ConvictionQueueHost.tsx</code>
Modal UI	<code>src/components/conviction/ConvictionFeedbackModal.tsx</code>
Storage	<code>src/lib/conviction/storage.ts</code>
Schools copy	<code>schoolsConvictionHeading()</code> etc. in <code>schoolsQuestRewardFlow.ts</code>

After Business island completion in Schools playthrough, learner is routed through conviction → profile → XP ladder → final challenge.

Profile updates

Concern	File(s)
Live snapshot hook	<code>src/hooks/useInvestorProfileSnapshot.ts</code>
Resolver	<code>src/lib/profile/resolveInvestorProfileSnapshot.ts</code>
Types	<code>src/lib/profile/investorProfileTypes.ts</code>
Dashboard UI	<code>src/components/profile/InvestorProfileDashboard.tsx</code>
Schools wrapper	<code>src/app/schools/profile/SchoolsProfileHub.tsx</code>

Profile reads **two sources**: engine `GameState` + Business framework localStorage (`iq-business-investor-framework:{companyId}`).

Persistence summary

Storage key / area	Contents
<code>investor-quest::state</code> (localStorage)	XP, completed quests, badges, pillar unlocks, active company
<code>iq-business-investor-framework:{companyId}</code>	Evidence ratings, section quiz passed flags, checklist progress
<code>sessionStorage</code>	Schools demo story flags, map brief dismiss, hub celebrate return, PWA hints
Supabase	Generated quest cards (when pipeline used) — not required for demo

Schools demo persists quest chain progress (unlike bank `/demo` in-memory mode). Reset via `SchoolsDemoHubResetButton` → `resetSchoolsDemoProgress.ts` .

6. Content model

Companies (playable)

File: `src/data/companies/index.ts`

Six companies with CIK, ticker, logo, tagline: `aapl` , `msft` , `tsla` , `nvda` , `nke` , `spot` .

Schools pick-company UI (`src/lib/schools/schoolsPickCompanyCatalog.ts`) shows **many more names** — most are **presentation-only** and not wired to `COMPANIES` .

Quest cards & learning content

Resolution priority (`src/lib/quests/questCardContentSource.ts`):

1. **Curated override** — `src/data/quests/content/{apple,nvidia,nike}.ts` (hand-polished)
2. **Database generated** — Supabase `company_quest_card_answers`
3. **Template fallback** — `plainEnglishAnswer: null` in templates (shows shimmer / pipeline pending)
4. **Generating** — in-flight pipeline state

Templates: `src/data/quests/templates/business.ts` (7 quests), plus `financials.ts` , `forces.ts` , `management.ts` .

Library binding: `src/data/quests/library.ts` — merges template + company + overrides.

Business quest slugs: `what-they-do`, `why-buying`, `everyday-life`, `how-it-works`, `why-they-stay`, `competition`, `who-competes` (see `src/lib/business/businessSlugMigration.ts` for legacy aliases).

Quiz questions

Content	File
Business per-quest quizzes	<code>src/data/quests/businessQuestQuizzes.ts</code>
Section checklist quizzes (5 MC each)	<code>src/lib/business/businessChecklistSectionQuizzes.ts</code>
Dynamic transforms	<code>src/lib/quests/dynamicQuizEngine.ts</code>
T/F eligibility rules	<code>src/lib/quests/quizTrueFalseEligibility.ts</code>
Order/ranking eligibility	<code>src/lib/quests/quizOrderEligibility.ts</code>
Token fill (<code>{Company.name}</code>)	<code>src/lib/quests/fillQuestTokens.ts</code>
Copy validation	<code>src/lib/quests/quizCopy.ts</code>

Business Island checklist content

Content	File
Section + principle definitions	<code>src/lib/business/businessInvestorFramework.ts</code>
Evidence card copy	<code>src/lib/business/businessInvestorEvidenceCards.ts</code>
Master principles ladder	<code>src/lib/business/masterInvestingPrinciples.ts</code>
Hub card models	<code>src/lib/business/buildBusinessHubCards.ts</code>
Quest marker positions	<code>src/lib/business/businessIslandQuestMarkerPositions.ts</code>

Map / island labels

Content	File
Schools Prodigy map layout	<code>src/lib/schools/schoolsProdigyMapConfig.ts</code>
Prodigy landmarks (SVG)	<code>src/components/schools/prodigyMap/ProdigyMapLandmarks.tsx</code>
Main game map hotspots	<code>src/lib/map/questMapDesktopConfig.ts</code>
Pillar screen images	<code>src/data/pillars/index.ts</code> + <code>src/lib/screenAssetUrIs.ts</code>

How to add/edit content

Edit a Business quiz question (section checkpoint):

1. Open `src/lib/business/businessChecklistSectionQuizzes.ts`
2. Edit the `questions` array for the section id (e.g. `company-overview`)

3. Run `npx tsc --noEmit` — dev build also asserts layout variety via `assertUniqueQuizLayouts` in `dynamicQuizEngine.ts`

Edit a per-quest island quiz:

1. `src/data/quests/businessQuestQuizzes.ts` for the slug key
2. Or company override in `src/data/quests/content/nvidia.ts` etc.

Edit quest reading cards:

1. Template structure: `src/data/quests/templates/business.ts`
2. Company-specific copy: `src/data/quests/content/*.ts`
3. For AI generation: admin `/admin/quest-generation` (needs Supabase + OpenAI)

Edit checklist sections/principles:

1. `src/lib/business/businessInvestorFramework.ts` — section defs, principle defs, emoji, quest slug mapping

7. Routing & screens

Schools presenter demo flow (polished — ****priority****)

Story steps from `src/lib/schools/schoolsDemoStoryMode.ts`:

Step	Route(s)	Status
<code>mission-brief-invitation</code>	<code>/schools/demo/mission-brief-invitation</code>	Polished — confidential invitation
<code>onboarding</code>	<code>/schools/screen5-onboarding</code> , <code>pick-interests</code> , <code>company-reveal</code> , <code>avatar</code>	Polished — multi-screen onboarding
<code>map-brief</code> / <code>map</code>	<code>/schools/map</code>	Polished — envelope brief + Prodigy SVG map
<code>business-island</code>	Map zoom (no separate page)	Polished — camera zoom into Business zone
<code>business-quest</code>	<code>/schools/business/[questSlug]</code>	Polished — reading + quiz + checklist
<code>conviction</code>	Modal overlay in playthrough	Polished
<code>profile</code>	<code>/schools/profile</code>	Polished (dashboard coded; some sub-routes stubbed)
<code>xp-ladder</code>	<code>/schools/xp-ladder</code>	Polished (re-exports main ladder)
<code>final-challenge</code>	<code>/schools/final-challenge</code>	Functional
<code>map-return</code>	<code>/schools/map</code>	Functional

Entry: `/schools` → redirects to `/schools/demo` (`src/app/schools/page.tsx`).

Schools side menu routes (`SCHOOLS\_DEMO\_MENU\_ITEMS` in `schoolsDemoMenu.ts`)

Route	Status
<code>/schools/map</code>	Polished (Prodigy)
<code>/schools/business</code>	Redirect → map hub zoom
<code>/schools/profile</code>	Polished
<code>/schools/pick-company</code>	Functional — catalog wider than playable companies

Business Island routes

Route	File	Status
<code>/schools/business/[questSlug]</code>	Re-exports <code>src/app/business/[questSlug]/page.tsx</code>	Primary Schools path
<code>/business</code>	<code>BusinessPageClient.tsx</code>	Canonical hub (desktop sidebar app)
<code>/business/[questSlug]</code>	<code>BusinessQuestClient.tsx</code> → <code>QuestDetailScreen</code> / <code>BusinessIslandQuestReading</code>	Full quest experience

Quiz screens

Quizzes are **not separate routes** — they render inside:

- `BusinessIslandQuestReading.tsx` (pillar quests + section quizzes)
- `QuestDetailScreen.tsx` (generic quest page)
- `BusinessChecklistSectionQuizFlow.tsx` (5-question section checkpoints)

Runner: `QuestQuizPanel.tsx` phases: `locked` → `ready` → `playing` → `summary`.

Profile

Route	Status
<code>/schools/profile</code>	Polished — <code>InvestorProfileDashboard</code> variant <code>schools</code>
<code>/profile</code>	Same dashboard, default chrome
<code>/profile/achievements</code>	Stub — "opening soon"
<code>/profile/sector-strength</code>	Stub
<code>/profile/industry-strength</code>	Stub

Map routes (two different maps)

Route	Map tech	Status
<code>/schools/map</code>	Prodigy SVG (<code>SchoolsProdigyMapScreen</code>)	Live Schools map
<code>/map</code>	Raster <code>desktop-map.png</code> (<code>QuestMapScene</code>)	Main app map — separate from Schools demo
<code>/schools/preview/map-*</code>	Style experiments (Roblox, Duolingo, Khan, ...)	Experimental only

Other islands (context only — ****not priority****)

Route	Status
<code>/schools/forces</code> , <code>/financials</code> , <code>/management</code>	Mirror main pillar pages; comment says "for now"
<code>/forces</code> , <code>/financials</code> , <code>/management</code>	Functional hubs with painted <code>.webp</code> scenery; less Schools-specific polish

Admin / ops (out of demo scope)

`/admin/quests` , `/admin/game-health` , `/admin/quest-generation` , `/dashboard/*` — internal tools.

Mobile preview

`/mobile-preview` , `/bank/mobile-preview` — embed previews for partner demos.

8. Assets

Central registry: `src/lib/screenAssetUrls.ts` (hub images, cache-bust `paintRev` query params).

Logos & brand

Asset	Path	Used in
Main logo	<code>public/logos/investor-quest-logo.png</code>	App shell, profile, preload in <code>layout.tsx</code>
Schools opening logo	<code>public/logos/schools-opening-logo.png</code> , <code>new-school-logo.png</code>	Schools intro screens
Company marks	<code>public/logos/companies/{aapl,msft,tsla,nvda,nke,spot}.svg</code>	Pickers, hub, profile
Extra company SVGs	<code>dkng</code> , <code>pltr</code> , <code>dis</code> , etc.	Pick-company UI only — not full game data

Map art

Asset	Path	Used in
Main app map	<code>public/logos/desktop-map.png</code>	<code>/map</code> (<code>QuestMapScene</code>)
Schools raster (legacy)	<code>public/logos/new-map.png</code>	<code>SchoolsQuestMapScreen</code> — preview/legacy
Prodigy map	<i>No raster — SVG in code</i>	<code>/schools/map</code> live

Island / hub scenery (AI-painted `.webp` — ****flagged****)

Asset	Path	Island
Business hub	<code>public/screens/biz-quest.webp</code>	<code>/business</code> , hub scenes
Financials	<code>public/screens/financial-quest.webp</code>	<code>/financials</code>
Forces	<code>public/screens/forces-quest.webp</code>	<code>/forces</code>
Management	<code>public/screens/management-quest.webp</code>	<code>/management</code>
Quest template backdrop	<code>public/screens/business-quest-template.webp</code>	Quest reading frame

Referenced via `BUSINESS_HUB_IMG_SRC` etc. in `screenAssetUrls.ts` .

Schools onboarding / identity (AI-painted — ****flagged****)

Asset	Path
Armor picker BG	<code>public/logos/choose-your-investor-armor.png</code>
Onboarding screens	<code>public/logos/screen2-onboarding.png</code> ... <code>screen5-onboarding.png</code> , <code>schools-sounds-like-you.png</code>
Identity photo	<code>public/images/schools/choose-identity.jpg</code>
Avatar portraits	<code>public/images/schools/avatars/*.png</code> (10 characters)
Mission brief art	<code>public/logos/schools-mission-brief.png</code> , <code>10k-mission-brief.png</code>

Avatar metadata: `src/lib/schools/avatars.ts` .

Profile visuals

Profile page is **coded UI** (`InvestorProfileDashboard.tsx`) — gradients, glass cards, live data. No single hero screenshot drives the page. Optional imagery: `public/images/schools/investing-mastery-hero.png` (may appear in ladder/marketing contexts).

Schools PWA

Asset	Path
Manifest	<code>public/schools/demo/manifest.webmanifest</code>
Apple touch icon	<code>public/schools/demo/apple-touch-icon.png</code>
Service worker	<code>public/sw-schools-demo.js</code> (versioned via <code>SCHOOLS_DEMO_SW_VERSION</code>)

Sounds

`public/sounds/schools-typewriter-key*.wav` — mission brief typing FX.

Referencing assets in code

- Prefer constants in `src/lib/screenAssetUrls.ts` for hub/map images
- Schools map config: `src/lib/schools/schoolsMapConfig.ts` (`SCHOOLS_MAP_IMAGE_SRC` for raster preview map)
- Next.js `Image` local patterns allow `/screens/**` and `/logos/**` (`next.config.ts`)

9. Current product state

Works well (polish these further, don't rewrite)

Area	Notes
Schools demo tour	Scripted flow from invitation → onboarding → map brief → Business zoom
Prodigy map	SVG overworld, mission envelope, Business island entry animation
Business Island hub	Quest markers, principles panel, investor checklist, journey progress
Quest reading	<code>BusinessIslandQuestReading</code> — cards, evidence flow, quality rating, section quizzes
Quiz variety	5 distinct layouts per section quiz via <code>dynamicQuizEngine</code> + <code>QuizQuestionView</code>
Quiz UX	Lock-in CTA, per-correct confetti, explanation callouts, Schools micro-XP
Profile dashboard	Live XP, level, company mastery, checklist rollup, quiz results — coded components
XP/progression	Reducer-backed; toasts and completion FX fire on quest pass
Investor checklist	Evidence cards, strong/weak ratings, section quiz gates, storage persistence

Still rough / inconsistent

Area	Notes
Mobile vs desktop	Business quest reading and quiz panels have mission-surface styles but spacing breaks on small phones need audit
Onboarding length	Many steps/screens before map — presenter flow is long; transitions could be tighter
Map brief replay	Session flags (<code>schoolsMapMissionBriefState.ts</code>) can confuse testers after refresh — reset button helps
Pick-company catalog	Shows companies without game data — misleading if user expects full quests
globals.css size	~28k lines mixing Schools, Business, admin, legacy — hard to navigate
ui/components explosion	Many effect components imported selectively; dead weight in bundle needs audit
Other three islands	Painted hub + template quests; no Schools-specific checklist/investor framework
Profile sub-pages	Achievements / sector strength are stubs
Explore lanes	<code>/explore/ai-frontier</code> , <code>market-giants</code> marked not ready

Quiz formatting (recent work)

- Section quizzes transform all-MC source into 5 formats: MC, T/F, fill-blank, odd-one-out, order/scenario
- Layout profiles enforced in dev (`quizLayoutProfiles.ts`)
- **Needs verification:** All 200+ seeds across sections pass layout uniqueness after future content edits

10. Known issues / limitations / fragile parts

Hard-coded / session fragile

- `SCHOOLS_DEMO_PRODUCTION_ORIGIN` — hard-coded Vercel URL
- Demo story step ↔ URL sync in `SchoolsDemoStoryOrchestrator.tsx` — refresh mid-flow can desync until flags reconcile
- `schoolsBusinessIslandZoomEnter.ts` — session flags for zoom/hub entered; easy to get stuck in hub phase without reset
- `BusinessIslandQuestReading.tsx` is **~1800 lines** — central god-component for quest flow
- Chunk load recovery via full page reload (`ClientAppRoot.tsx`) — masks deploy cache issues

Duplication

- Three route trees: `/schools/` , `/schools/demo/` , `/schools/preview/` (+ canonical `/business` , `/demo`)
- Map implementations: Prodigy SVG vs raster `SchoolsQuestMapScreen` vs main `QuestMapScene`
- XP constants split between `xpEconomy.ts` (engine) and `schoolsQuestRewardFlow.ts` (Schools UI)
- Dual persistence: engine state + `businessInvestorFrameworkStorage` — profile must read both

Placeholder / incomplete

- Quest templates with `plainEnglishAnswer: null` — AI pipeline fill
- `schoolsPickCompanyCatalog.ts` companies without `COMPANIES` entries
- Profile achievements/sector routes — stubs
- Root `tmp-zone-*.jpg` — dev calibration artifacts (should not ship)
- Filename typos in assets: `choose-your-avitar-bg.png` , `eports.png` — suggests WIP

Visually inconsistent

- Schools **mission cream/gold** surface (`PillarQuestTheme.cardChrome === "mission"`) vs dark sci-fi default island theme
- Business hub mixes **coded SVG scene** + **painted webp** background depending on layout branch
- Large generated `ui/*` components vs bespoke `iq-*` CSS — two design dialects

Likely to break if touched carelessly

- `dynamicQuizEngine.ts` + `quizLayoutProfiles.ts` — dev throws on duplicate layouts
- `persistence.ts` migrations — `STATE_VERSION` bumps must stay backward compatible
- `schoolsDemoStoryMode.ts` route mapping — breaking pathname breaks presenter tour
- Hydration: no `Math.random()` on first paint rule (see `.cursor/rules/investor-quest-experience.mdc`) — quiz shuffles happen in click handlers / effects only

Confusing for new developers

- `/schools/business` redirects to map — hub is **on the map**, not a separate island page
- `QuestDetailScreen` vs `BusinessIslandQuestReading` — Schools business uses the latter inside quest client
- "Investor Quality Check" vs "Investor Checklist" vs "Master Investing Principles" — three parallel progression UX layers on same quests

11. Recommended polish tasks

Must polish first

- ☐ **Business Island hub UI** — marker clarity, checklist panel density, mobile tap targets (`SchoolsBusinessHubIslandLayout.tsx` , `BusinessIslandQuestMarkers.tsx`)
- ☐ **Quest card layout** — reading flow in `BusinessIslandQuestReading.tsx` + `BusinessQuestTemplateFrame.tsx` ; reduce scroll fatigue on mobile
- ☐ **Quiz format variety** — verify all section quizzes show 5 distinct layouts; polish mission-surface styles in `QuizQuestionView.tsx` + `globals.css` (`iq-quiz-*`)
- ☐ **Profile page** — dashboard is coded; replace any remaining static marketing images; tighten Schools variant spacing (`InvestorProfileDashboard.tsx` , `SchoolsProfileHub.tsx`)
- ☐ **XP updates after quiz** — confirm micro-XP + quest XP + section quiz XP all surface in profile immediately (`useInvestorProfileSnapshot` , reducer events, toast timing)
- ☐ **Map flow / mission brief** — envelope sequence clarity, replay rules after reset, Business zoom guide label (`SchoolsMapPageClient.tsx` , `MissionEnvelopeBriefSequence.tsx`)

- ☐ **Desktop + mobile responsiveness** — audit `BusinessIslandQuestReading` , `QuestQuizPanel` , map HUD, profile grid at 320px–1440px
- ☐ **Onboarding / intro** — shorten or skip paths for returning users; smoother transitions between story steps
- ☐ **Reset/demo reliability** — `SchoolsDemoHubResetButton` + `resetSchoolsDemoProgress.ts` should always yield predictable fresh tour
- ☐ **globals.css hygiene** — group Schools/Business quiz styles; document key `iq-*` classes used in demo

Nice to have later

- ☐ **Forces / Risks Island** — Schools-specific polish, checklist framework parity with Business
- ☐ **Financials Island** — same
- ☐ **Management Island** — same
- ☐ **Advanced investing checklist** — extend framework beyond Business sections
- ☐ **More companies** — wire pick-catalog entries to `COMPANIES` + content overrides
- ☐ **Backend / admin** — Supabase auth, remote progression store, quest generation at scale
- ☐ **Preview map lab** — promote one alternate map style or delete unused preview routes
- ☐ **Bundle cleanup** — tree-shake unused `src/components/ui/*` effects
- ☐ **Capacitor** — TestFlight / Play Store packaging for Schools PWA wrapper

12. Developer notes

Conventions worth following

- **Game mutations** only through `useGame().actions` — never mutate `GameState` in components
- **Schools presenter URLs** use `/schools/demo/*` — `schoolsDemoHref.ts` resolves learner vs demo paths
- **Port 3003 only** for dev — see `.cursor/rules/dev-server-port.mdc`
- **Pointer events** on decorative map/FX layers must be `none` — interactive nodes above (see experience rules)
- **Commits** — user prefers explicit ask before git commit; deploy is via `main` push

Testing the core loop quickly

1. `npm run dev`
2. Sidebar → **Schools Live Demo**
3. Complete onboarding → map brief → tap Business island
4. Open `what-they-do` quest → read cards → take quiz → quality check
5. Open checklist section quiz (after evidence) — confirm 5 different quiz layouts
6. Visit `/schools/profile` — confirm XP/checklist updated
7. Use hub reset button if state feels stuck

Typecheck before PR

```
npx tsc --noEmit
npm run lint
```

Needs verification

- Full `.env.local` variable list for your machine (file is gitignored)
- Which Supabase migrations are applied in production Vercel env
- Exact Vercel project dashboard URL / team access for new developer
- Whether `OPENAI_API_KEY` is set in production (only needed for live quest generation, not static demo)
- Performance budget on low-end mobile for `globals.css` size and Framer Motion on map zoom

Related docs in repo

Doc	Topic
<code>README.md</code>	Local dev quick start + live URLs
<code>src/ARCHITECTURE.md</code>	Engine layer design
<code>docs/CAPACITOR.md</code>	Mobile wrapper + production URL
<code>.cursor/rules/investor-quest-experience.mdc</code>	Product feel, animation, interaction rules
<code>.cursor/rules/dev-server-port.mdc</code>	Port 3003, game health, fast quest dev

Generated for developer handoff — focus: Schools demo + Business Island. Last aligned to `main` branch post quiz-layout variety + investor checklist work.

PDF export: Run `npm run generate:handoff-pdf` to regenerate `HANDOFF.pdf` from this file (cover page, table of contents, syntax-highlighted code, page numbers).